

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

In the xy -plane, the y -coordinate of the y -intercept of the graph of the function f is c . Which of the following must be equal to c ?

Answer

- A. $f(0)$
- B. $f(1)$
- C. $f(2)$
- D. $f(3)$

Correct Answer: A

Rationale

Choice A is correct. A y -intercept is the point in the xy -plane where the graph of the function crosses the y -axis, which is where $x = 0$. It's given that the y -coordinate of the y -intercept of the graph of function f is c . It follows that the coordinate pair representing the y -intercept must be $(0,c)$. Therefore, c must equal $f(0)$.

Choices B, C, and D are incorrect because $f(1)$, $f(2)$, and $f(3)$ would represent the y -value of the coordinate where $x = 1$, $x = 2$, and $x = 3$, respectively.